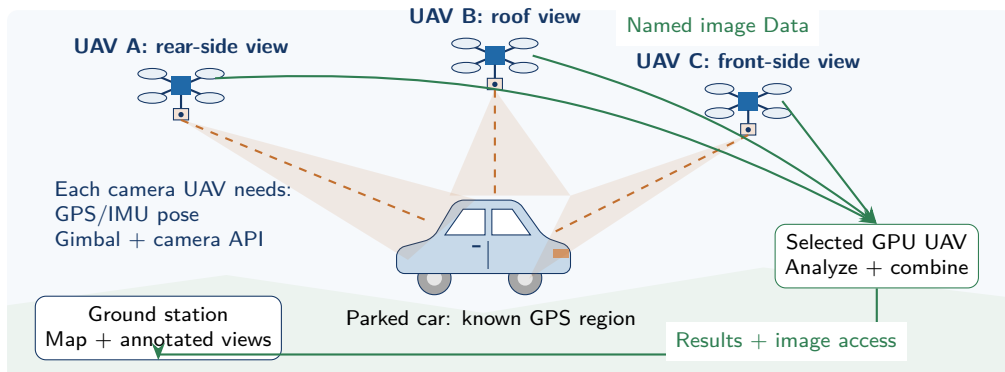
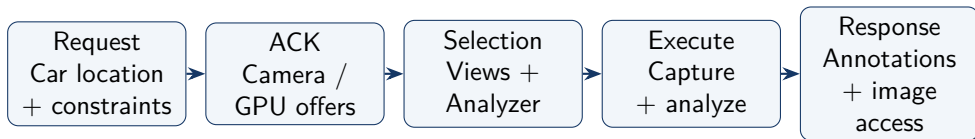


Real-World Scenario: Inspect a Parked Car



Evidence: Illustrative field scenario, not a completed flight experiment. Orange: camera view. Green: application data flow, not flight paths.

NDNSF Connects Roles; Sensors Must Deliver the Views



- ▶ **NDNSF**: discover authorized Providers, assign roles, retrieve protected data and track completion.
- ▶ **UAV application**: navigate, point the camera, capture and validate the observation.
- ▶ **Perception**: detect the car, associate the same car across views and combine evidence.

Key point

Successful message exchange does not prove that a UAV captured the right target.

Evidence: Proposed inspection workflow; component/simulation evidence does not establish field qualification.

Three Gaps Between Simulation and the Field

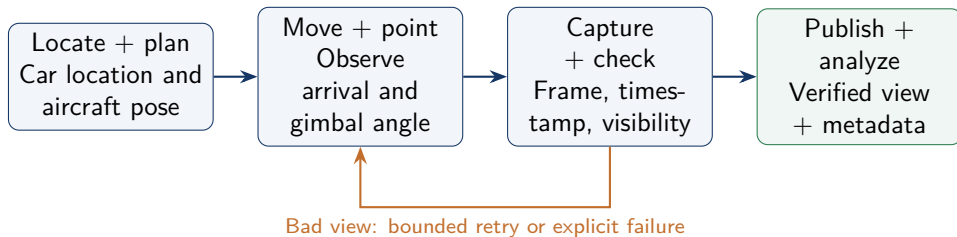
Gap	Simplified example	Real deployment needs
Position / pointing	Known coordinates; assigned views	GPS + IMU, camera/gimbal calibration, coordinate frames, synchronized pose and exposure
Useful images	Car assumed visible	Real control/capture APIs, feedback, blur/occlusion checks and bounded retries
Multi-view analysis	Images supplied to a detector	Same-target association, uncertainty-aware fusion and independent ground truth

Key point

Known GPS narrows the search; it does not guarantee visibility or make three detections a reliable multi-view result.

Evidence: Candidate tools: MAVSDK control/telemetry; OpenCV calibration/geometry. Hardware support and integration remain to be verified.

Close One Physical Loop Before Adding More UAVs



- ▶ A control ACK is not capture success: check the image and measured camera pose.
- ▶ A pixel needs geometry/depth for world localization; views need correspondence before fusion.
- ▶ Keep stabilization and safety rejection onboard, independent of network delays.

Evidence: Proposed feedback loop: reject stale pose, missing frames and exhausted deadlines.

A Short Route to a Credible Field Demonstration

1. **Bench:** one real camera/gimbal; verify commands, capture, calibration and pose/time alignment.
2. **Flight simulation:** use the same adapter; inject navigation error, occlusion and link loss. Use PX4 software/hardware-in-the-loop where supported.
3. **One UAV:** inspect a parked car at a surveyed location; measure pointing error, useful-view rate and latency.
4. **Multiple UAVs:** compare one versus complementary views; measure recognition quality, time/energy cost and participant failures.

Key point

Next milestone: one real, correctly aimed image with synchronized pose and independently checked target visibility.

Evidence: Planned stages, not completed experiments. Start with a parked car; moving-car tracking is outside this first demonstration.

Available Hardware: Programmable Gimbal Cameras

SIYI A8 mini

- ▶ Integrated camera + three-axis gimbal.
- ▶ SDK control over Ethernet UDP or UART; Ethernet video output.
- ▶ Candidate for the first camera-control bench experiment.

Gremsy Pixy family

- ▶ Gimbals for compatible camera payloads.
- ▶ SDK/MAVLink control; camera support depends on the model.
- ▶ Candidate when a specific imaging payload is required.

Verify before selection: angle commands, attitude feedback, image access, payload weight, power and mounting.

Key point

Pointing at a car's GPS location still needs aircraft pose and camera calibration; a controllable gimbal supplies only part of the loop.

Evidence: SIYI A8 mini manual; Gremsy SDK. Candidate devices, not purchased or validated hardware.